

Introduction to Python Programming

Welcome to the Introduction to Python Programming lecture. In this session, we will explore the fundamental concepts of programming using Python, including understanding source code, syntax, the role of the Python interpreter, writing simple expressions, and how Python processes these commands.

1. Understanding Source Code

- **Source Code:** This is the sequence of instructions written in a programming language that a computer can execute. In Python, source code is written in plain text files with a .py extension.
- Example of a simple source code:

```
print('Hello, World!')
```

2. Syntax in Python

- **Syntax:** This refers to the set of rules that define the structure of statements in a programming language. Each programming language has its own syntax, and using incorrect syntax will result in errors.
- In Python, the syntax for printing a message is:

```
print('Your message here')
```

3. The Role of the Python Interpreter

- **Python Interpreter:** Python is an interpreted language, meaning that it directly executes the instructions written in the source code without the need for prior compilation. The interpreter translates the code into machine instructions that the computer can understand.
- The Python interpreter also provides an interactive environment where you can type commands and see immediate results. For example:

```
```
```

```
3 + 7 10 print('Hello') Hello ```
```

## 4. Writing Simple Expressions

- **Expressions:** An expression is a combination of values and operators that Python can evaluate to produce a result. For example,  $1 + 4 * 3$  is an expression that evaluates to 13 due to operator precedence.
- **Operators:** In Python, you can perform various arithmetic operations using operators:
  - + for addition
  - - for subtraction
  - \* for multiplication
  - / for division
  - % for modulus (remainder)
  - \*\* for exponentiation

## 5. Operator Precedence

- Python follows specific rules for the order of operations, known as precedence. For example, multiplication and division are performed before addition and subtraction. Parentheses can be used to alter this order.
- Example:
  - $1 + 3 * 4$  evaluates to 13.
  - $(1 + 3) * 4$  evaluates to 16.

## 6. Using the Python Console

- The Python console is where you can execute Python commands interactively. You can also write scripts in a text editor and run them in the console. For example:

```
age = input('How old are you? ') print('Your age is', age)
```

## 7. Variables and Data Types

- **Variables:** A variable is a named piece of memory that can store a value. You can assign a value to a variable using the assignment statement, e.g., `x = 5`.
- In Python, you can store different data types in variables, such as integers, floats, strings, and more.
- Example of variable assignment:

```
gpa = 3.14
```

## 8. Input and Output

- **Input:** You can read user input using the `input` function. The result can be stored in a variable for further processing.
- **Output:** The `print` function is used to display messages or results to the console.
- Example:

```
name = input('What is your name? ') print('Hello,', name)
```

## 9. Conclusion

In this lecture, we covered the basics of programming using Python. We discussed source code, syntax, the Python interpreter, writing simple expressions, and how Python processes commands. These concepts form the foundation for further learning in Python programming, and I encourage you to practice by writing your own Python code and experimenting with the interpreter. Thank you for your attention!

## Quiz Questions

**Q1: What file extension is used for Python source code?**

- .txt
- .java
- .py
- .html

*Correct: .py*

**Q2: What is the purpose of the Python interpreter?**

- To compile Python code into machine language
- To execute Python code directly
- To store Python files
- To edit Python source code

*Correct: To execute Python code directly*

**Q3: Which of the following is an arithmetic operator in Python?**

- &&
- ||

- +

- !

*Correct: +*